

MODEL PAPER

TED (15) 6111

Reg No:.....

REVISION-2015

Signature:.....

**SIXTH SEMESTER DIPLOMA EXAMINATION IN TOOL AND DIE
ENGINEERING**

MODERN PRODUCTION PROCESS

[Time: 3 hours

(Maximum marks: 100)

Marks

PART – A

(Maximum marks: 10)

I. (Answer the following questions in one or two sentences. Each question carries 2 marks)

1. Classify robots.
2. Name any four types of non-conventional machining methods.
3. Define G-code.
4. Write any four advantages of EDM.
5. Define CAM.

(5x2=10)

PART – B

(Maximum marks: 30)

II. (Answer any *five* of the following questions. Each question carries 6 marks.)

1. Differentiate between DNC and CNC.
2. Define M-code and explain various M-codes.
3. Reproduce the detailed view of a Ultrasonic Machining.
4. Explain about NC system.
5. List the different types robots.
6. Illustrate main features of CMM.
7. Explain CAD hardware and CAD software.

(5 X 6 = 30)

P.T.O

PART – C

(Answer one full question from each unit. Each full question carries 15 marks.)

UNIT – I

- III. a. Describe the principles of Electric Discharge Machining. (8)
b. Explain Electrical Discharge Grinding. (7)
OR
- IV. a. Describe Ultrasonic Machining. (8)
b. Illustrate Electro Chemical Machining. (7)

UNIT – II

- V. a. Explain Flexible Manufacturing System. (8)
b. Describe the different methods of control in CNC machines. (7)
OR
- VI. a. State the principles of Numerical control machine tools. (8)
b. Compare Open loop system and Closed loop system. (7)

UNIT – III

- VII. a. Illustrate programming methods in CNC system. (8)
b. Explain the uses of subroutines. (7)

OR

- VIII. a. Explain the use of following and G-codes. (8)
g02, g01, g05, g03, g90, g91, g33, g94
b. Describe circular and linear interpolation. (7)

UNIT-IV

- IX. a. Explain the different types of manipulators in robots. (8)
b. Identify the industrial application of robots. (7)
OR
- X. a. Describe Computer integrated manufacturing (CIM) system. (7)
b. State the functions and benefits of CAM. (8)